MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift

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Code and artifacts: https://github.com/RudraChopra/mosaic-certified-release/tree/main

Abstract

AI systems increasingly expose small decision interfaces, such as routing, moderation, eligibility, or triage signals derived from richer internal representations. A weak probe on the internal state does not guarantee that the released signal is safe, especially after population shift. We introduce MOSAIC, a software-only layer that releases a task-specific stochastic token only when two finite-sample contracts hold: Bayes-optimal source inference within each task label is bounded, and worst-stratum task error remains below a registered threshold. The guarantee covers every target law supported by labeled target audit data; otherwise MOSAIC abstains. The key device is a simultaneous confidence set on categorical laws constructed before channel optimization. One event therefore covers every stochastic channel searched on the same table. A bridge linear program represents supported target shift as a common source-blind transform plus bounded source-specific residuals. We prove exact attacker and utility envelopes, global finite-alphabet optimization, matching sample-complexity bounds, bounded transcript accounting, and anytime-valid recertification. In 1,000 locked safety trials, naive same-table selection releases contract-violating channels on 42.1% of tables; MOSAIC records none. The primary five-benchmark study releases 20/100 interfaces, all on Bias-Bios, with 0/20 held-out violations. Three separately locked natural-shift confirmations in text, census, and images release 75/120 interfaces with 0/75 held-out violations. A disclosed post-hoc stress test shows why the broader bridge class matters: among 36 direct target-table releases, MOSAIC blocks all 16 interfaces that fail on an admitted target law.

1 Introduction

Routing, moderation, eligibility, and triage systems often expose a compact decision signal derived from a richer model state. Representation editing is commonly evaluated by fitting a probe to that state (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), although probe design can change the apparent conclusion (Pimentel et al. 2020). This leaves two deployment gaps: a different downstream rule may recover the attribute, and a release selected on a noisy validation table may fail after population shift. The operational object that needs a contract is the public interface, not the claim that every trace of a concept vanished internally.

We study a task-specific finite-token interface. A tokenizer fixed without the certification fold maps an internal representation to $C \in [K]$; a source-blind stochastic channel $M \in \mathrm{Stoch}(K, L)$ releases the only public token Z. Within each task label, the contract bounds the balanced accuracy of the Bayes-optimal source attacker rather than a chosen probe. This isolates conditional source signal from label prevalence. Pre-release shift is a common Markov transform plus bounded source-specific contamination, permitting large source-blind drift while charging the part that can create new source differences.

MOSAIC, Minimax-Optimized Source-Agnostic Invariant Channels, is a general certified-release layer instantiated here for edited representations. Its input is a fine token C derived from the model state; its output is a smaller token Z and a task decoder. It first uses simultaneous multinomial confidence regions to describe what is known about C. It then uses labeled target audit data to learn which population shifts are supported. Finally, it searches for a stochastic channel whose source distinguishability and task error satisfy both contracts throughout that supported shift class. If no such channel exists, it returns ABSTAIN.

The central object is the data-certified bridge: a finite-sample test that turns labeled target observations into a population class on which the chosen interface remains valid. The upstream confidence event then covers the whole post-channel family, so privacy and utility can be optimized jointly without testing each channel. This differs from finite-candidate risk control because the event lies before the channel search. Locked experiments compare this transport certificate with direct target-table certification, plug-in and held-out selection, LTT, deterministic release, official erasers, and an exact population oracle.

Pipeline in one pass. The practitioner fixes the tokenizer, contracts, and data splits before using the certification folds. MOSAIC then (1) builds confidence regions for the reference and target token laws, (2) fits the largest common source-blind target transform supported by both regions, (3) globally optimizes the release channel and task decoder against the resulting worst-case laws, and (4) recomputes every bound before releasing one persistent token. Any failed step returns

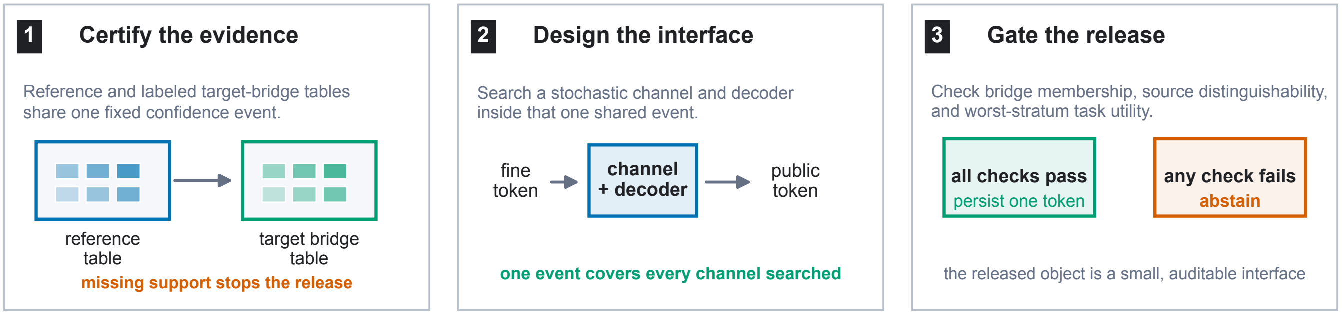

Figure 1: MOSAIC groups the release decision into three stages: certify the reference and labeled target evidence, design a source-blind stochastic token on that shared confidence event, and release only after bridge membership, source distinguishability, and task utility all pass. Otherwise it abstains.

ABSTAIN. Figure 1 groups these operations into three stages.

2 Related Work

Concept-erasure methods remove source-associated directions, covariance, or components (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), while leakage audits show why a weak probe is not a universal guarantee (Pimentel et al. 2020; Elazar et al. 2021; Chowdhury et al. 2025). FARE certifies every downstream classifier for restricted finite encoders, and Fair Normalizing Flows bounds strong adversaries through tractable densities (Jovanovic et al. 2023; Balunović, Ruoss, and Vechev 2022). Stochastic preprocessing and fair representations motivate randomized release channels (Calmon et al. 2017; Makhdoumi et al. 2014; Agarwal and Deshpande 2024).

LTT, conformal risk control, and Pareto testing certify registered decisions and may abstain (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023); robust validation addresses declared uncertainty sets (Tibshirani et al. 2019; Cauchois et al. 2024). MOSAIC differs by certifying the categorical experiment before channel search, covering an uncountable channel family without separate tests. Its source-blind transform is a Blackwell garbling and its residual is a contamination neighborhood (Blackwell 1953; Le Cam 1964; Torgersen 1991; Huber 1964). The bridge makes that shift model testable from target data, whereas unrestricted shift is impossible to certify (Ben-David et al. 2010; Klivans, Stavropoulos, and Vasilyan 2024; Chandrasekaran et al. 2025).

3 Contract and Running Example

Let $S \in [G]$ denote source, $Y \in [J]$ task label, and $C \in [K]$ the fine token. For a fixed label, write $p_{s,y}(c) = \Pr(C = c \mid S = s, Y = y)$. The source-blind release channel M produces $Z \in [L]$. The balanced accuracy of the strongest downstream source attacker within label y is

$$A_y(P, M) = \frac{1}{G} \sum_{z=1}^{L} \max_{s \in [G]} (p_{s,y} M)(z), \tag{1}$$

with normalized advantage $\mathrm{Adv}_G(A) = (GA-1)/(G-1)$. This is chance-normalized and ranges from zero to one. The registered source contract is $\max_y \mathrm{Adv}_G(A_y) \le \tau_P$.

Running case. We follow one registered BiasBios-Clinical interface throughout the paper. Binary gender is the audited source and a clinical-profession decision is the task. After rank-four INLP, a fixed task score maps each biography to one of four fine tokens. MOSAIC releases two tokens with decoder $g(z) = z$ and

$$M = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ .725 & .275 \\ 0 & 1 \end{pmatrix}.$$

It never exposes the edited embedding. The source-advantage bounds are .201 and .204 within the two task labels, and the certified worst-stratum task error is .261, clearing the registered .35 and .40 contracts. The untouched diagnostic values are .052 and .184.

C	fine token	Z	public token
M	channel	g	task decoder
$p_{s,y}$	reference law	$q_{s,y}$	target law
$t_y, T_y, r_{s,y}$	bridge terms	τ_P, τ_U	contracts

The main contract concerns source inference within a task label. A joint-law extension covers natural-mixture source inference: define $X = (Y, C)$, estimate $P(X \mid S)$ without balancing labels, and lift the channel as $\widetilde{M}((y,c), z) = M(c, z)$. The same envelope then certifies inference from Z; using output (Y, Z) covers an attacker given the true label as well.

For each supported source–label stratum, the certification table supplies an empirical law $\widehat{p}_{s,y}$ and radius $\epsilon_{s,y}$. The simultaneous confidence set is $\mathcal{C}_{s,y} = \{p \in \Delta_K : \|p - \widehat{p}_{s,y}\|_1 \le \epsilon_{s,y}\}$. Under conditionally i.i.d. categorical sampling within each registered source–label stratum, the simultaneous event

$$\mathcal{E} = \bigcap_{s,y} \{\|\widehat{p}_{s,y} - p_{s,y}\|_1 \le \epsilon_{s,y}\} \tag{2}$$

has probability at least $1 - \delta$ after a fixed allocation across strata. We use the Weissman multinomial radius (Weissman et al. 2003) in the confirmation, although any valid simultaneous region can replace it. The tokenizer and fine alphabet must be fixed without these observations.

The external law for label y belongs to the structured class

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \qquad t_y \geq 1 - \eta_y, \tag{3}$$

where T_y is a source-independent fine-token channel from a registered convex uncertainty set and each residual r_s is arbitrary. Both components pass through the same release channel M. This placement matters: a constant-row software channel cannot leak information that appears before it.

4 Why Same-Table Optimization Is Valid

For a bounded vector v, define the exact confidence-set support function

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p \in \Delta_K : \|p - \widehat{p}\|_1 \leq \epsilon} p^\top v. \tag{4}$$

It is obtained by moving at most $\epsilon/2$ mass from low to high coordinates of v, or from an equivalent linear-program dual. For attacker assignment $a \in [G]^L$, let $v_{M,a,s}(c) = \sum_z M(c,z)\mathbf{1}\{a(z) = s\}$. We then define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{5}$$

Theorem 1 (Adaptive exact envelope)**.** *On $\mathcal{E}$, simultaneously for every row-stochastic M, including a channel selected as an arbitrary function of the same table, $A(P, M) \leq \overline{A}(\widehat{P}, M)$. For fixed M, the right-hand side is the exact supremum over the product of the stated ℓ_1 confidence sets.*

The proof writes the Bayes attacker as a maximum over deterministic assignments. For a fixed assignment the source confidence sets separate, giving one support function per source; all maxima are finite and commute. The implication is pointwise over the complete stochastic-channel space, which is why adaptive selection needs no channel-count correction. In the running case, the same confidence event covers the displayed M and every alternative four-by-two channel considered by the optimizer.

5 Learning the Supported Shift Class

MOSAIC certifies structured-shift membership from labeled target bridge data. Let a labeled bridge sample from the target population define simultaneous confidence sets $\mathcal{D}_{s,y}$ for its fine-token laws $q_{s,y}$. Write

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \qquad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \tag{6}$$

For each label, the bridge program chooses retained mass t and $W = tT$ by solving

$$\begin{aligned} \max_{t,W} \quad & t \\ \text{s.t.} \quad & W \geq 0, \quad W\mathbf{1} = t\mathbf{1}, \quad 0 \leq t \leq 1, \\ & h_{s,y}(W_{:z}) \leq \underline{q}_{s,y}(z) \quad \forall s, z. \end{aligned} \tag{7}$$

The support functions and coordinate lower bounds have exact LP duals, so Eq. (7) is one linear program rather than a transform search.

Running case, continued. For the released BiasBios interface, the bridge LP certifies retained common mass .799 and .796 in the two task labels. MOSAIC therefore charges source-specific residuals of .201 and .204 in its downstream attacker and utility envelopes. These are population bounds from separate reference and bridge confidence regions, not fitted similarities on the diagnostic fold. The resulting channel merges the first two score bins, randomizes only the third, and preserves the fourth: this is the complete public decision surface an auditor must inspect.

Theorem 2 (Adaptive finite-sample bridge)**.** *Suppose the reference and bridge confidence sets jointly cover every $p_{s,y}$ and $q_{s,y}$. Any feasible $(\widehat{t}, \widehat{W})$ with $\widehat{t} > 0$, including the same-table maximizer, defines $\widehat{T} = \widehat{W}/\widehat{t}$ and residual laws satisfying*

$$q_{s,y} = \widehat{t}\, p_{s,y} \widehat{T} + (1 - \widehat{t}) r_{s,y} \quad \forall s. \tag{8}$$

Moreover, Eq. (7) returns the largest retained mass certified uniformly over the product confidence region.

Indeed, feasibility and coverage imply $q_{s,y}(z) \geq p_{s,y}^\top \widehat{W}_{:z}$ coordinatewise. The difference is nonnegative and sums to $1 - \widehat{t}$, so normalization constructs $r_{s,y}$. Conversely, robust membership requires exactly these coordinatewise inequalities; maximizing the reference support and minimizing the bridge coordinate yields Eq. (7). Thus the learned singleton transform $\{\widehat{T}_y\}$ and contamination $\widehat{\eta}_y = 1 - \widehat{t}_y$ can be inserted into the certificates below on the same event. If a required bridge source–label stratum is absent, its confidence set is the full simplex, every coordinate lower bound is zero, and Eq. (7) forces $\widehat{t} = 0$. For the practitioner, the bridge either supplies a measured residual budget for the target population or stops the release before a channel is deployed.

6 Certifying Source and Task Risk after Shift

We use three names consistently: the *data-certified bridge* tests shift membership, the *transform-exact certificate* enumerates the learned shift geometry, and the *capacity fallback* upper-bounds it when that geometry cannot be enumerated.

Define the arbitrary differential-shift capacity

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{9}$$

This equals $\sup_{r_1,\ldots,r_G} A(R, M)$. For $G = 2$, its normalized value is the Dobrushin coefficient, the largest total variation distance between two rows of M.

For a transform polytope with registered extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define the shifted envelope

$$\begin{aligned} \overline{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y,\, a \in [G]^L} \sum_s \Big[& \\ (1 - \eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) & \\ + \eta_y \max_c v_{M,a,s}(c) \Big]. & \end{aligned} \tag{10}$$

Theorem 3 (Transform-exact structured-shift certificate)**.** *On $\mathcal{E}$, every same-table selected M and every external law in Eq. (3) satisfy $A_y(Q, M) \leq \overline{A}_y^X(M)$. For fixed M, the* *right-hand side is the exact supremum over the stated confidence sets, transform polytope, retained masses, and arbitrary residual laws.*

For fixed T, retained mass, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are respectively $\Phi(\widehat{p}, \epsilon; Tv)$ and $\max_c v(c)$. The latter is no smaller, so the worst retained mass is $1 - \eta_y$. The support objective is convex in T; its maximum over a polytope occurs at an extreme. These finite maxima commute, proving exactness. Since the statement holds pointwise for every M, it survives adaptive channel selection.

When transform extremes cannot be enumerated, a capacity-fallback corollary is available. Define $\rho_T(M) = \max_c \mathrm{TV}((TM)(c,\cdot), M(c,\cdot))$ and $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$. Then

$$A(Q, M) \leq (1 - \eta_y) B_y(M) + \eta_y \,\mathrm{Cap}_G(M). \tag{11}$$

The exact envelope in Eq. (10) is never larger. When $\eta_y = 0$ and $TM = M$, both reduce to the reference certificate; when $\eta_y = 1$, both reduce to exact distribution-free capacity.

Because the fallback separately upper-bounds every law in the same structured class, both transform-exact source and utility values are pointwise no larger than their fallback values. "Never worse" is therefore an implementation check implied by the theorem; experiments measure how often tightness changes the release decision. Proofs are in the supplement.

For decoder $g : [L] \to [J]$, let $u_y(c) = \sum_z M(c,z)\mathbf{1}\{g(z) \neq y\}$. The matching exact conditional error certificate is

$$\overline{E}^X_{s,y}(M, g) = \max_{T \in \mathcal{V}_y} \Big[(1 - \eta_y) \\ \Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tu_y) + \eta_y \max_c u_y(c)\Big]. \tag{12}$$

It is the exact supremum over the same confidence and shift sets. Thus M, g, within-label source distinguishability, and worst-stratum utility can all be selected on one event.

> **Guarantee at a glance.** With probability at least $1 - \delta$ over the registered reference and bridge samples, every released (M, g) meets both contracts for every target law in the learned bridge class. If the bridge has zero retained mass, a required stratum is missing, or no channel satisfies both bounds, MOSAIC abstains.

Statistical and session extensions. For a distribution-uniform ℓ_1 region that must cover every bounded functional selected from the same K-category table, the minimax resolution rate is $n = \Theta((K + \log(1/\delta))/\gamma^2)$; the Weissman construction matches it up to constants (Han, Jiao, and Weissman 2015). For a registered session of r possibly correlated fine items, compiling the mechanisms into the joint transcript channel gives the same exact envelope. A scalable capacity bound is $1 - \prod_{i=1}^{r}(1 - \alpha_i)$, where α_i is the Dobrushin coefficient of item channel i. Finally, replacing fixed-time regions by Dirichlet-mixture e-process regions makes every recertification time and optional stopping rule valid on one event (Ryu and Wornell 2024). The supplement proves these statements, gives a finite-cover extension for continuous release classes, and an all-threshold DKW corollary.

Unrestricted source-specific shift can be erased only by a fine-token-independent channel, so a bounded residual is a deployment assumption rather than a hidden technical convenience. Likewise, a missing required source stratum is unestimable rather than evidence of safety; both boundary proofs are in the supplement. A constant-row channel is the $\varepsilon = 0$ endpoint of local differential privacy (Warner 1965).

7 Release Algorithm

For fixed decoder, support functions, transformed scores, and residual maxima have linear epigraphs. Enforcing every transform extreme and attacker assignment gives one LP in the KL channel entries; enumerating J^L decoders yields the global finite-alphabet optimum. Acceptance requires solver optimality, valid primal residuals, and independent certificate recomputation.

Release protocol. Fix tokenizers, alphabets, thresholds, split roles, and confidence allocation before viewing certification or bridge folds. Construct simultaneous regions, solve Eq. (7), and enumerate decoder LPs subject to every source constraint. Replay each returned certificate, then select the feasible registered tokenizer with minimum certified error under a fixed tie break. Zero retained mass, missing support, no feasible row, or a failed numerical check returns ABSTAIN; otherwise only (M, g) and Z are released.

With $R = \max_y |\mathcal{V}_y|$, the direct formulation is polynomial in K for fixed (G, J, L, R) and exponential only in the released alphabet L. Constraint generation adds the strongest omitted attacker until none violates the contract; the supplement gives the proof and scaling details.

8 Evaluation Design

Questions and error budgets. The evaluation is organized around three questions. *Safety* asks whether same-table selection or an unsupported shift can release a violating channel. *Retention* asks how often a safe interface can still be released. *Scope* asks which populations and interface sizes the available data can support. Every study is locked before its own outcomes and uses a separate familywise $\delta = .05$ allocation. The paper does not combine those allocations into one study-wide probability statement.

Synthetic safety and power. The primary simulation uses 1,000 independent certification tables per cell, two source–label populations, two shift regimes, eight release rules, and up to five sample sizes. A population evaluator that is not used for selection grades every chosen channel. The key comparisons are naive same-table optimization, MOSAIC, held-out certification, finite-family LTT, and deterministic release. A second 12,000-table study tests bridge membership as its labeled target sample grows from 500 to 5,000 observations per source–label stratum. It includes one valid common transform and two registered violations: a source-specific transform and contamination beyond the declared budget.

Primary five-benchmark study. We register 100 jobs, with 20 seeds on each of Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, BiasBios-Clinical, and GaitPDB.

Each job uses 13 frozen edits from official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations. Construction data fit the task score and four-bin tokenizer. Separate observations estimate the reference law, labeled target bridge data fit the supported shift class, and an untouched target fold supplies diagnostics. Neither diagnostics nor their outcomes enter selection. The primary contracts are source advantage $\tau_P = .35$ and worst-stratum error $\tau_U = .40$; threshold sweeps use $\tau_U \in \{.30, .35, .40, .45, .49\}$. These values are study contracts, not universal recommendations for deployment.

The bridge requires target S and Y audit labels because it estimates $q_{s,y}$. Those labels are used only inside the audit and are not released. When a required stratum is absent, MOSAIC abstains. A separate proxy-label study tests the alternative of inferring S and also reports abstention when proxy uncertainty is too large.

Natural-shift breadth. Two additional confirmations test geography and image origin. The ACS study registers 60 transfers from California to four target states, spanning three tasks and five seeds (Ding et al. 2021); whole geographic units separate bridge and diagnostic folds. The CINIC-10 study treats CIFAR-10 versus ImageNet origin as source and uses disjoint construction, reference, bridge, and diagnostic folds (Darlow et al. 2018). A disclosed sequential extension uses 40 fixed seeds and a stricter familywise allocation. Complete split, manifest, amendment, and replay details are in the supplement.

9 Results: Safety, Retention, and Scope

Safety: Does Certification Block Invalid Releases?

Figure 2 gives the primary synthetic result. Naive same-table optimization releases a population-violating channel on 421/1,000 tables at the safety boundary. MOSAIC releases none of those channels. Across 8,000 MOSAIC certification tables, every accepted release satisfies the exact population contract.

The primary five-benchmark study shows the same pattern across progressively weaker release rules. At $\tau_U = .40$, strict MOSAIC releases 20/100 jobs with 0/20 held-out violations. Removing confidence regions but retaining the fitted bridge releases 47 jobs and violates 7/47 estimable diagnostics. Ignoring the target-membership gate releases 80 jobs and violates 18/60 estimable diagnostics, while also releasing 20 Camelyon jobs whose required source stratum is absent. Unconditional deployment violates 38/80 estimable jobs. Thus each removed safeguard changes an observed decision: intervals prevent plug-in errors, the bridge gate rejects unsupported populations, and abstention prevents contract failures.

The bridge-membership study asks a different safety question: does the target population belong to the declared shift class? A valid common transform is accepted on 0%, 31%, 100%, and 100% of tables as labeled target support increases through 500, 1,000, 2,000, and 5,000 observations per stratum. Two registered violations, one source-specific transform and one excess contamination law, are rejected on every table. These are deliberately clear violations; local power immediately outside the bridge boundary is not measured by this experiment.

K	Rule	Releases	Held-out violations
4	MOSAIC	15/60	0/15
4	Direct table	47/60	0/47
8	MOSAIC	0/60	–
8	Direct table	47/60	0/47

Table 1: Natural California-to-state transfer across 60 registered ACS jobs. The held-out diagnostic uses complete geographic units that do not enter bridge fitting or channel selection.

Retention: What Does the Guarantee Cost?

Safety is useful only if the method can still release. In the synthetic retention cell, continuum MOSAIC releases 579/1,000 tables, compared with 283/1,000 for Holm LTT over the same 500 fixed channel–decoder pairs. Transform-exact certification further improves retention by using the learned shift geometry: at $n = 125$ it releases 53.0% versus 0.3% for the generic capacity fallback, and at $n = 250$ the rates are 99.9% and 57.8%. Every accepted channel in these comparisons satisfies the population contract.

Real-data retention is more concentrated. All 20 primary releases come from BiasBios-Clinical, where median certified retained mass is .797; the other four benchmarks abstain because of missing support or an infeasible joint source–utility contract. One released rank-four INLP interface maps four fine tokens to two persistent public tokens. Its certified source-advantage and worst-stratum error bounds are $.204$ and $.261$, versus untouched diagnostic values $.052$ and $.184$. The gap between certified and diagnostic values is the finite-sample and shift margin paid for the guarantee.

Natural geographic transfer. The locked ACS study contains 60 jobs: three tasks, four target states, and five seeds. At the primary $.40$ error contract, $K = 4$ MOSAIC releases 15/60 jobs with 0/15 held-out population diagnostic violations. Releases cover all four target states and two tasks. The direct target-table rule releases 47/60 with 0/47 held-out violations. At $K = 8$, MOSAIC abstains on all 60 jobs while the direct rule again releases 47/60. The larger token table therefore raises the confidence cost enough to eliminate MOSAIC retention at the available sample size.

Natural-shift breadth and interface utility. Across three separately locked natural-shift confirmations, MOSAIC releases 75/120 jobs (62.5%) with 0/75 held-out violations: 20/20 BiasBios-Clinical, 15/60 ACS geographic, and 40/40 CINIC-10 image-origin jobs. The minimum leave-one-domain-out pooled release rate is 43.8%. Relative to each selected four-token task interface, the median channel cost is 0.60 points on ACS and 0.60 points on CINIC-10; the maximum costs are 1.42 and 2.86 points. The supplement gives every seed and the full comparison boundary.

The ACS comparison makes the retention cost explicit. At $K = 4$, direct target-table certification releases 47/60 jobs while MOSAIC releases 15/60; neither rule has a held-out population diagnostic violation. The direct rule certifies

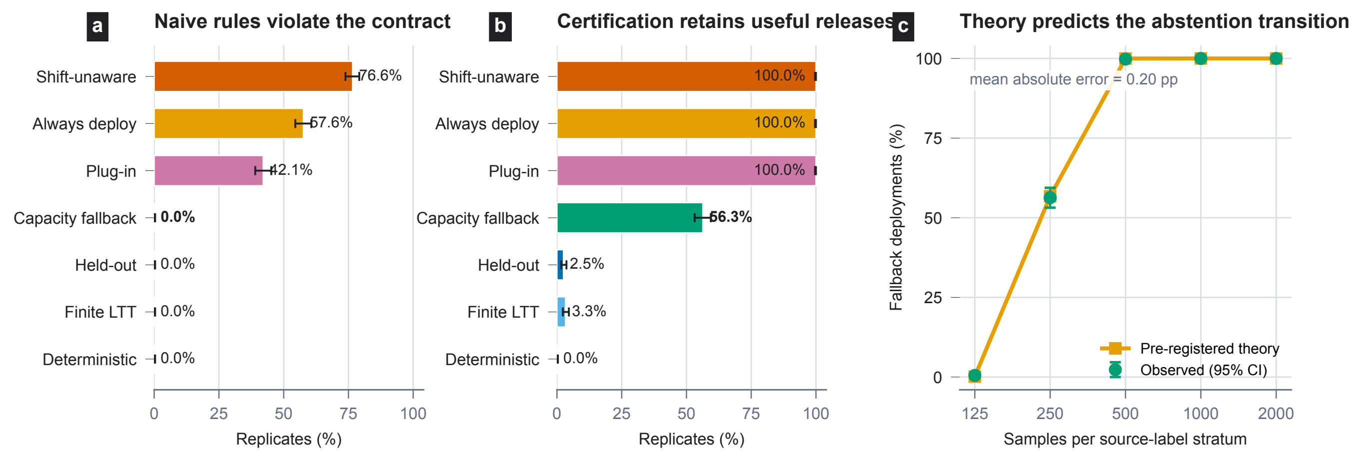

Figure 2: Hash-locked confirmation with 1,000 independent certification tables per cell. Error bars are exact 95% Clopper–Pearson intervals. (a) Naive same-table selection falsely accepts 42.1% of boundary tables; MOSAIC records no false acceptance. (b) In the retention cell, MOSAIC's capacity fallback releases 56.3%, versus 2.5% for held-out certification, 3.3% for finite-family LTT, and 0% for deterministic release. (c) Release frequency follows the preregistered sample-size curve with 0.20 percentage-point mean absolute error.

the law represented by one target table. MOSAIC certifies every law in the larger bridge class supported by reference and target evidence. No natural panel in this study shows a held-out contract failure for the direct rule, so the natural evidence establishes successful MOSAIC transport and its retention cost, not a higher natural failure frequency for direct certification.

Scope: When Is the Broader Bridge Needed?

The direct and bridge rules protect different target-law sets. In the primary study, the direct rule releases 36/100 jobs with 0/36 diagnostic violations, versus 20/100 and 0/20 for MOSAIC. To isolate the set difference, a disclosed post-hoc stress test maximizes source inference over each direct release's stored bridge class. Sixteen of the 36 direct interfaces violate their contract on an admitted law; MOSAIC releases the other 20 and abstains on all 16 failures. Every stress law is an exact bridge-class member outside the direct confidence region. This result proves that the broader class can contain harmful laws; it does not estimate how often those laws occur naturally.

The reverse mismatch is also detectable. In 8,000 locked tables from two bridge-invalid populations, the true target law lies inside the direct target region on 7,983 tables, while MOSAIC's membership gate abstains on all 8,000. A direct certificate can therefore be valid for a sampled target law even when that population is outside MOSAIC's declared transport model.

Alphabet size creates a second boundary. Four released tokens improve only five of 100 primary jobs relative to two tokens. Synthetic optimization remains subsecond through $K = 64$, $G = 4$, and $L = 4$, but attacker enumeration is exponential in released alphabet size L; these experiments do not establish scaling to large public alphabets. In the original ACS $K = 8$ study, MOSAIC abstains on every job. Increasing certification support from 6,000 to 16,000 rows per stratum later enables two of three registered employment transfers, showing that this boundary is statistical as well as computational.

Finally, the contract covers one persistent token per item. A registered bounded session can be certified as one joint transcript, but repeated fresh draws or added side information require a new joint certificate. Likewise, the real proxy-label study abstains because source-label uncertainty is too large. Current deployment therefore requires confidential target source labels for auditing; inferring those labels does not automatically preserve power. Complete threshold frontiers, per-seed tables, exact-rational replays, proxy results, and amendment history are in the supplement.

10 Discussion

Which rule should a practitioner use? If deployment is tied to one well-sampled target population, direct target-table certification is simpler and retained more interfaces in our ACS study. MOSAIC is designed for a different promise: one interface must remain valid throughout a larger, data-supported class of target laws. The bridge quantifies that class and stops deployment when the available target evidence cannot support it. Retraining can improve a target predictor, but it does not provide this release-interface guarantee.

One decision from start to finish. In the running BiasBios case, the reference table describes the four fine-token laws, while the separate target audit table supports common retained masses .799 and .796 in the two task labels. MOSAIC treats the remaining .201 and .204 as arbitrary source-specific residuals. It then searches every four-to-two stochastic channel and binary decoder, selecting the displayed M. Replaying the exact envelopes gives source advantage at most .204 and worst-stratum error at most .261. Both are below the registered .35 and .40 thresholds, so the system releases one persistent token and its decoder. If a required target stratum were missing, the bridge retained mass would be zero and the same pipeline would return ABSTAIN.

Certificate versus diagnostic. The certified values $.204$ and $.261$ are not predictions of average observed performance. They are worst-case bounds over sampling uncertainty and every law in the learned shift class. The untouched diagnostic values, $.052$ and $.184$, describe one realized hold-out population. Their difference measures the price of finite-sample uncertainty and transport coverage. Diagnostics evaluate a selection after the fact; only the certificate controls the declared law class before release.

How to read a contract. The values $.35$ source advantage and $.40$ worst-stratum error are registered experimental thresholds, not recommended defaults. An application owner must set τ_P and τ_U from its own harm and utility requirements before certification. The probability $1-\delta$ refers to the random reference and bridge samples within one registered study. It is not the fraction of future people protected, and separate studies do not silently share one error budget.

Operational boundary. The runtime emits one stored token or ABSTAIN and fails closed on missing source–label support, an expired audit, excessive proxy-label uncertainty, or an infeasible contract. The guarantee does not cover the edited embedding, hidden system state, timing information, or side information not included in the declared attacker view. A bounded session can be compiled into one joint transcript certificate. Unbounded fresh draws require a new mechanism or a new joint contract because source evidence can accumulate.

The theorem chain in plain language. The three main theorems answer separate questions. Theorem 1 says channel search cannot invalidate the statistical event because that event was built for the fine-token law before any channel was selected. Theorem 2 says the bridge inequalities are enough to write every covered target law as a common source-blind transform plus residuals. Theorem 3 then evaluates the strongest source attacker and largest task error over exactly that class. Together they turn "this channel looked good on validation" into "this channel meets both contracts for every law supported by the audit evidence."

What is fixed and what is learned. Before certification, the user fixes the tokenizer, source and task strata, attacker view, thresholds, split roles, and confidence allocation. The data then determine the confidence regions, bridge transform, residual budget, release channel, and decoder. Target diagnostics remain untouched until the release decision is complete. This separation prevents the evaluation fold from quietly becoming part of the design process.

Question	Certified answer
Attacker	Bayes-optimal source inference from the declared public view
Task risk	Maximum conditional error over registered source–label strata
Target scope	Every law in the data-certified bridge class
Randomness	Registered reference and labeled target audit samples
Released object	One persistent stochastic token and its task decoder
No support	ABSTAIN

Current empirical envelope. The real-data studies use binary sources and public alphabets of two or four tokens. Synthetic scaling reaches $K = 64$, $G = 4$, and $L = 4$, while the natural $K = 8$ ACS study shows that larger internal token tables can require much more audit data. The bridge misspecification study verifies two registered failure modes but does not map power for every near-boundary violation. These boundaries define the next empirical tests rather than changing the stated certificate.

What MOSAIC contributes. Unlike finite-family LTT, MOSAIC certifies the categorical experiment before release, so every same-table channel and decoder inherits one pointwise envelope. The bridge then converts labeled target evidence into a testable shift class. The result is a release decision whose guarantee, retention cost, and failure boundary are all explicit.

11 Conclusion

MOSAIC turns an edited representation into a small public interface with a testable deployment contract. Its upstream confidence event permits global same-table channel optimization, while the bridge converts labeled target data into a supported shift class. The system releases when source and task contracts hold throughout that class and abstains when the evidence is insufficient. The experiments show both sides of that decision: certification blocks unsafe selection, and broader transport coverage can reduce retention.

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